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LAB : 06

Code –

AI LAB 06

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```python
print("Tic Tac Toe Lab 06")
```

```python
board = [
    ['x','o','x'],
    ['o','x','_'],
    ['_','_','o']
]

for row in board:
    print(row)
```

```python
def evaluate(board):

    # Check rows
    for row in board:
        if row[0] == row[1] == row[2]:
            if row[0] == 'x':
                return 10
            elif row[0] == 'o':
                return -10

    # Check columns
    for col in range(3):
        if board[0][col] == board[1][col] == board[2][col]:
            if board[0][col] == 'x':
                return 10
            elif board[0][col] == 'o':
                return -10

    # Check diagonals
    if board[0][0] == board[1][1] == board[2][2]:
        if board[0][0] == 'x':
            return 10
        elif board[0][0] == 'o':
            return -10

    if board[0][2] == board[1][1] == board[2][0]:
        if board[0][2] == 'x':
            return 10
        elif board[0][2] == 'o':
```

AI LAB 06

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[1]

```
        return 0
```

[ ]

```
    result = evaluate(board)
    print("Board Value:", result)
```

[2]

```
!pip install pyvirtualdisplay
import tkinter as tk
from pyvirtualdisplay import Display

display = Display(visible=0, size=(800, 600))
display.start()

def click(row,col):
    global player

    if buttons[row][col]['text']=="":
        buttons[row][col]['text']=player

        if player=="x":
            player="o"
        else:
            player="x"

root=tk.Tk()
root.title("Tic Tac Toe")

player="x"
buttons=[[0,0,0],[0,0,0],[0,0,0]]

for r in range(3):
    for c in range(3):
        buttons[r][c]=tk.Button(root,text="",width=10,height=4,
        command=lambda r=r,c=c:click(r,c))

        buttons[r][c].grid(row=r,column=c)

root.mainloop()

display.stop()
```

AI LAB 06

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[ ]

```
    print("Board Value:", result)
```

[2]

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root.mainloop()

display.stop()
```

```
Requirement already satisfied: pyvirtualdisplay in /usr/local/lib/python3.12/dist-packages (3.0)
```